Orbital Dragons to Hunt Cryptids: Neural Networks, Quantum-Inspired Data Science, and the Busy Beaver Problem

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The busy beaver value $BB(n)$ is the longest finite runtime of any n-state Turing machine started on a blank tape. $BB(5)$ was settled in 2024 with a computer-checked proof. $BB(6)$ is blocked by a handful of machines whose tapes secretly run Collatz-style arithmetic; nobody can prove whether they stop. This paper studies two of these machines and surveys the rest. The approach is empirical first: the machine's behavior is logged as a sequence of events, the event data embedded in logarithmic, simplex, and hyperbolic coordinates, and the dynamics found to be piecewise affine in a five-integer state. The description is then proven correct in the Lean proof assistant, down to the machine's individual steps and with a minimal axiom footprint. The result is a machine-checked theorem: if a simple five-integer recurrence never enters an explicit finite set of bad configurations, the machine never halts. The converse pressure is also proven: entering a bad configuration forces the machine to halt. The open problem is thereby moved, without loss, from a Turing machine to a short arithmetic recurrence. The renewal map of the recurrence is then derived in closed form and shown to have a width-five fatal window around the diagonal, and the two mechanical proof routes with computational evidence: no congruence invariant up to modulus 384 excludes the window, and the branch dynamics has exponential factor complexity, so it is not conjugate to a rotation. Measuring the recurrence over its whole seed space shows the set of dying seeds has natural density zero at the Borel–Cantelli rate ($\approx 12/X$), so the studied machines' seeds are typical survivors. Surveying the entire holdout set with a two-ended harness (forward orbit against backward halt cone, regimes fitted in between) classifies all 1,094 open six-state machines: zero halts, zero alignments; four fixed-dimension machines, exactly the certified ones; an eleven-machine *array class* whose state spaces grow in dimension, for whose leading member an exact word-rewrite calculus is constructed (a two-phase odometer with a block-trisection boundary rule) validated against 2×10^8 raw steps; and 307 single-law machines among 346 with regime structure. The program is framed on a verification spectrum whose proven endpoint is $\mathsf{MIP}^* = \mathsf{RE}$: kernel checking at the bottom, certified predictive emulation (this paper) in the middle, and entangled-prover verification of halting-type claims as the goal. Every empirical claim is validated against 10^9 raw machine steps, and the recurrence is tested for 10^8 iterations. For the theory of learning, the pipeline is representation learning carried to a proof, and it is offered as a benchmark: unbounded, exactly labeled, law-governed sequence data on which a learned representation has a provable ground truth, with a two-prover game that verifies the forecasts of any learned emulator at constant cost per check.

The central certified statement, one per machine:

Theorem 1. *Checked by the Lean kernel at axioms* $\{\text{propext}, \text{Quot.sound}\}$*:* $\mathsf{SafeMetaOrbit} \to \neg\mathsf{RawHalts}(\mathsf{blankStart})$*, and configurations outside the guarded region halt within about fifty steps, so* $\mathsf{SafeMetaOrbit}$ *is the halting problem restated as arithmetic.*

References

[1] The bbchallenge Collaboration. Determination of the fifth Busy Beaver value. arXiv:2509.12337.

[2] The bbchallenge Collaboration. Turing machine deciders, part I. arXiv:2504.20563.

[3] P. Michel. The Busy Beaver Competition: a historical survey. arXiv:0906.3749.

[4] S. Aaronson. The Busy Beaver Frontier. *SIGACT News* 51(3), 2020.

[5] T. Tao. Almost all orbits of the Collatz map attain almost bounded values. arXiv:1909.03562.

[6] C. Moore. Generalized shifts: unpredictability and undecidability in dynamical systems. *Nonlinearity* 4 (1991).

[7] P. Kůrka. On topological dynamics of Turing machines. *Theoret. Comput. Sci.* 174 (1997).

[8] O. Viro. Dequantization of real algebraic geometry on logarithmic paper. arXiv:math/0005163.

[9] G. Litvinov. The Maslov dequantization, idempotent and tropical mathematics. arXiv:math/0507014.

[10] M. Nickel and D. Kiela. Poincaré embeddings for learning hierarchical representations. NeurIPS 2017.

[11] C. Series. The modular surface and continued fractions. *J. London Math. Soc.* 31 (1985).

[12] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen. $\mathsf{MIP}^* = \mathsf{RE}$. arXiv:2001.04383; *Comm. ACM* 64(11), 2021.

[13] H. Mousavi, S. Nezhadi, and H. Yuen. Nonlocal games, compression theorems, and the arithmetical hierarchy. STOC 2022; arXiv:2110.04651.

[14] T. Tao et al. The Equational Theories Project. arXiv:2512.07087.

[15] Palomar: a registry of Lean-verified mathematics, 2026. `palomar-registry.org`.